

1. Draw a Data Flow Diagram up to level 2 for Library Management System and Account Management System.
2. Draw an E-R diagram for the following scenario:
Consider a bus ticketing system that records information about passenger, bus and route. Passenger is assign to a bus to travel on route. A bus contains many passengers and passenger can be assigned to only one bus. Many buses travel in same route but a bus can travel in one route. The attribute of passenger are pid, gender, phone; similarly bus has reg, color whereas route contains rid, distance, rate.
3. Draw a class Diagram for the given scenario.
Office contains two department web department and mobile app department, each department consist of five to eight employees. Employees consist of manager and works in both department.
4. Construct a Class diagram of the Banking management system.
5. Draw a Use Case diagram for ATM System.
6. Draw an Activity Diagram for an ATM System where
 - a. Customer is withdrawing the balance
 - b. Checking the balance
7. Draw an Activity Diagram for the scenario in which customer is ordering the food of his/her choice using online websites.
8. Draw a sequence diagram of ATM System showing cash withdrawing process.
9. Draw a Sequence for the following scenario:
A customer request for some product to order. The system checks whether the product is available in stock or not in stock. Stick system reply if the product is available. After the product is ready, customer should pay the amount of that product. System should update the information and he/she receive the receipt.
10. Draw a Deployment Diagram for any web application and desktop application.

Process to make Lab report:

- i. Cover Page
- ii. Table of content
- iii. Content according to table of content. Include introduction, explain symbols used in each diagram and steps to create each diagram (handwritten) and draw the diagram using app and print and insert it in the report. For e.g.

let me clarify question number 3.

- Introduction of Activity diagram
- Explain symbols used in activity diagram.
 - i. Start
 - ii. Activity
 - iii. Decision
 - iv. Control Flow
 - v. Synchronization bar
 - a) Fork
 - b) Join
 - vi. End State
- Steps to create activity diagram.
 - i. Identify the initial and final states
 - ii. Identify the intermediate activities needed,
 - iii. Identify the conditions or constraints.

iv. Draw the diagram with Appropriate Notations: